MSc THESIS · EXPERIMENT DESIGN

The forecast-error experiments

How one experiment is structured, how we build a case, how forecast errors enter, which bids we compare, and the pipeline that delivers the results.

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OVERVIEW

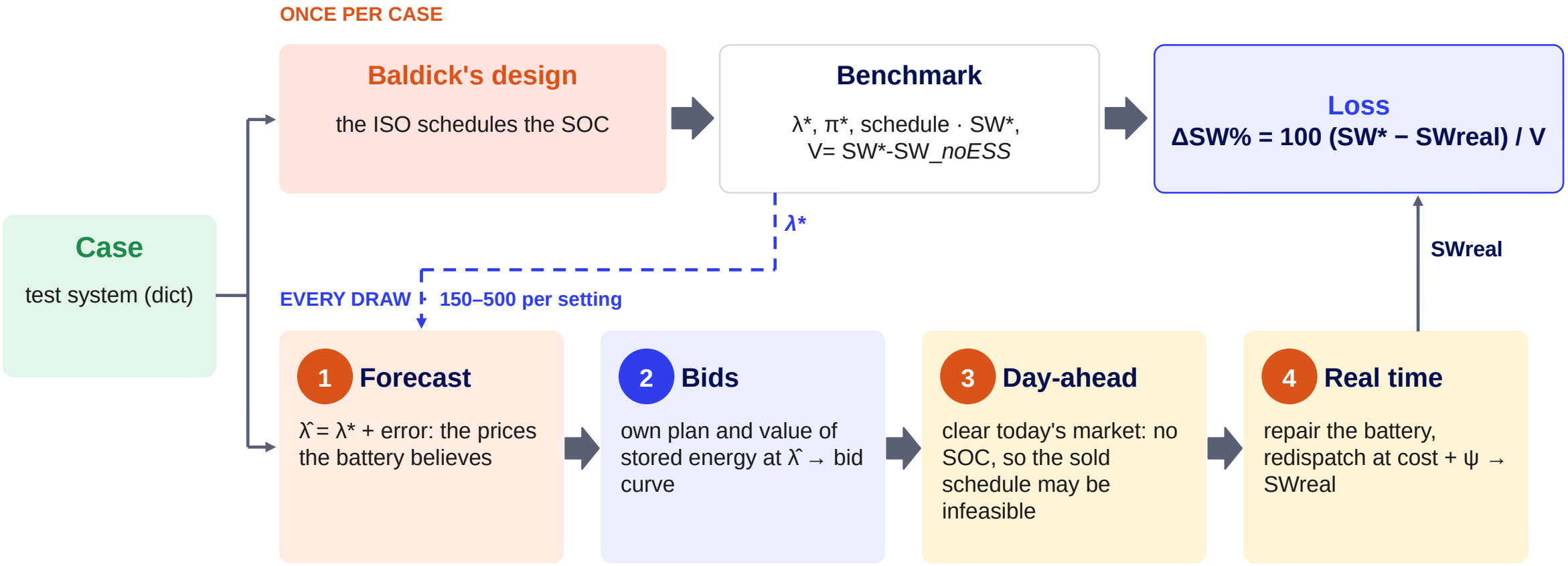
1	**Structure**	One experiment: a benchmark solved once, a market run per draw	slide 3
2	**The case**	One dictionary of generators, demand and batteries; every model is built from it	slide 4
3	**Forecast errors**	How the error is built ($\hat{\lambda}$, z, χ, η) and why these values of σ, ρ, b, ω	slides 5–6
4	**Bid families**	B★, B★p, Bfcst, Bpriv: each isolates one piece of information	slide 7
5	**The pipeline**	Final deliverable: explorer and albums, one draw step by step (σ, bias, ρ)	slides 8–11



Parts 1–4 build the experiment; part 5 is what it delivers.

One experiment: a benchmark solved once, a market run per draw

1 · STRUCTURE



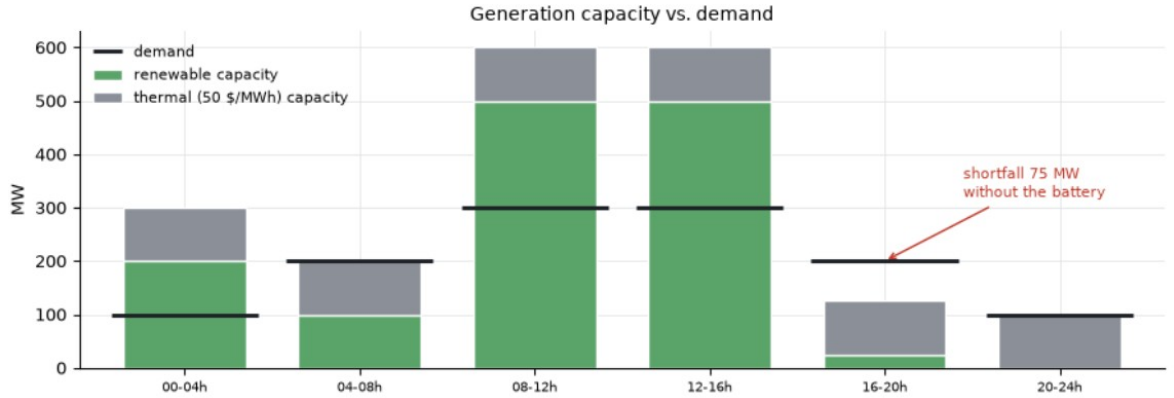
Only step 1 sees the error: steps 3–4 and the welfare always use the true system.

Same yardstick for both sides: the only difference is what the battery knew when it bid.

How we build a case: one dictionary, every model built from it

2 · THE CASE

```
case = dict(
  T=6, dT=4.0,            # six 4-hour intervals
  gens=[renewable, thermal], # cost, capacity
  D=[[100,200,300,300,200,100]], # MW
  ess=[battery],          # θ and w
  N=1, lines=[], loads=[]) # one bus
```



Capacity against demand: without storage, 16–20h is 75 MW short.

Generators	renewable: cost 0, capacity 200/100/500/500/25/0 MW · thermal: 50 $/MWh, 100 MW
Demand	inelastic, 100–300 MW; shedding priced at VOLL = 1000 $/MWh
Battery	α = 0.9, β = γ = 1, ε = ζ = 1 $/MWh, 0–800 MWh, starts empty, w = 40 $/MWh

Benchmark (once)

Baldick's design on the case gives λ* = 43.1 · 50 · 0 · 0 · 50 · 50 $/MWh, π*, the schedule, SW* and V.

Next cases, same format

Stage 2: a fleet of 5–20 batteries.
Stage 3: a reduced RTS-GMLC area, many days.

Small and fully checkable first: every number of this case can be verified by hand.

How forecast errors are implemented

$$\hat{\lambda}_{e,t} = \max\left(\underline{\lambda},\quad \lambda^{\star}_{t} + \underbrace{b\,\bar{\lambda}}_{\text{bias}} + \underbrace{\sigma\, z_{e,t}}_{\text{random}}\right)$$

$$z_{e,t} = \sqrt{\omega}\ \underbrace{\chi_t}_{\text{common}} + \sqrt{1-\omega}\ \underbrace{\eta_{e,t}}_{\text{own}}$$

Both pieces follow the same AR(1) rule · all shocks and starting values ~ N(0, 1)

$$\chi_t = \rho\,\chi_{t-1} + \sqrt{1-\rho^2}\,w_t \qquad \text{common: one sequence for all batteries}$$

$$\eta_{e,t} = \rho\,\eta_{e,t-1} + \sqrt{1-\rho^2}\,v_{e,t} \qquad \text{own: one sequence per battery } e$$

σ
size, % of the mean price

ρ
persistence hour to hour

b
bias (level error)

ω
share common to all batteries

1 — Seed from the draw number
seed(d) = SeedSequence(base_seed, d): any draw can be regenerated

2 — Same shape in every setting
σ and b only scale and shift it; ρ and ω only recombine it

3 — Floor at 0
a forecast never predicts prices this market cannot produce

4 — Inserted in one place
λ̂ replaces λ* only in the battery's own plan and value π

Notation

$e,\ t$	battery, interval (4 h)	$z_{e,t}$	random error, variance 1
$\lambda^{\star}_{t}$	true price (benchmark)	χ_t	common piece, all batteries
$\hat{\lambda}_{e,t}$	forecast of battery e	$\eta_{e,t}$	own piece of battery e
$\bar{\lambda}$	mean of λ* over the day	$w_t,\ v_{e,t}$	independent N(0, 1) shocks
$\underline{\lambda}$	price floor (0 $/MWh)	σ, ρ, b, ω	the four knobs (left)

Four knobs on one shared error shape: differences between settings come from the knob alone.

Why these error values

3 · FORECAST ERRORS

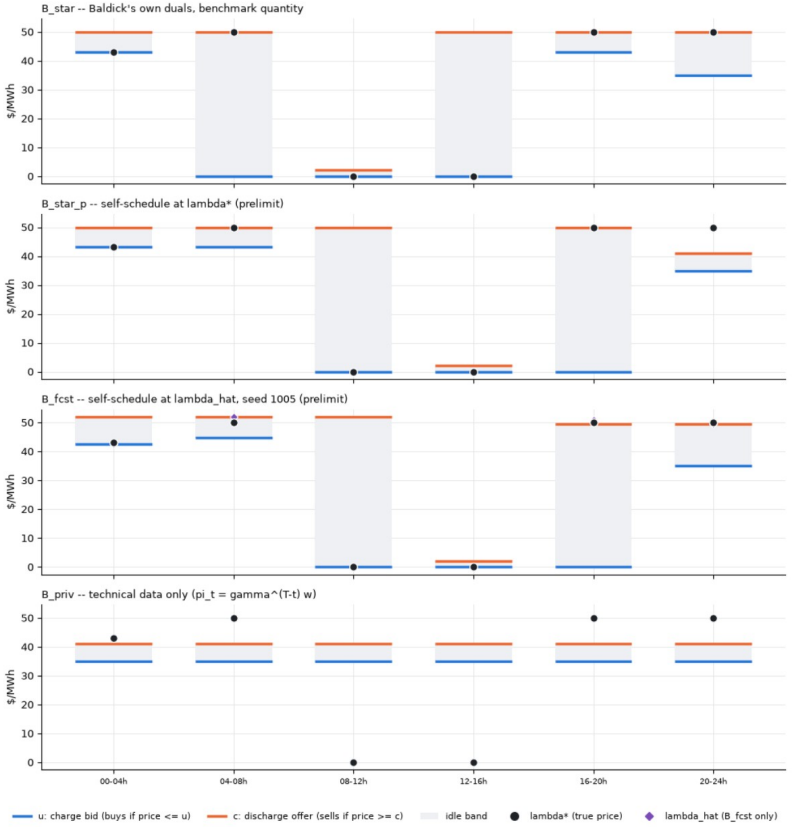
Knob	Values	Why
σ size	0, 2.5, 5, 10, 20, 40% of $\bar{\lambda}$	Published day-ahead forecasts (Lago et al. 2021): state of the art ≈ 6–18%, naive ≈ 15–47%. 0 is the control; 40% means no skill.
ρ persistence	0, 0.5, 0.9	Wind forecast errors in the project's RTS-GMLC data: autocorrelation 0.49 at 4 h (our interval), 0.87 at 1 h.
b bias	−10, 0, +10% of $\bar{\lambda}$	A systematic level error of the same size as the best random error; symmetric, so sign effects show.
ω common share	0, 0.5, 1	Own forecaster vs one shared forecast (vendor or TSO); matters only with several batteries.
floor	0 $/MWh	This market never clears below 0 (free renewables); a parameter, since real markets do.
ψ premium	0, 10, 50 $/MWh	Frictionless · close to real imbalance penalties · scarcity stress test. To agree with the supervisors.

Every published value sits inside the grid; calibration later says where reality is.

Four bid families, each isolating one piece of information

4 · BID FAMILIES

	Prices from	Quantity	What it isolates
B★	true prices, Baldick's duals	Baldick's schedule	**gate: must lose 0**
B★p	true prices, own selection	own plan at λ*	**cost without quantity plan**
Bfcst	forecast λ̂	own plan at λ̂	**main experiment: + price forecast**
Bpriv	none: technical data only	full capacity	**lower bound: is a forecast worth it?**

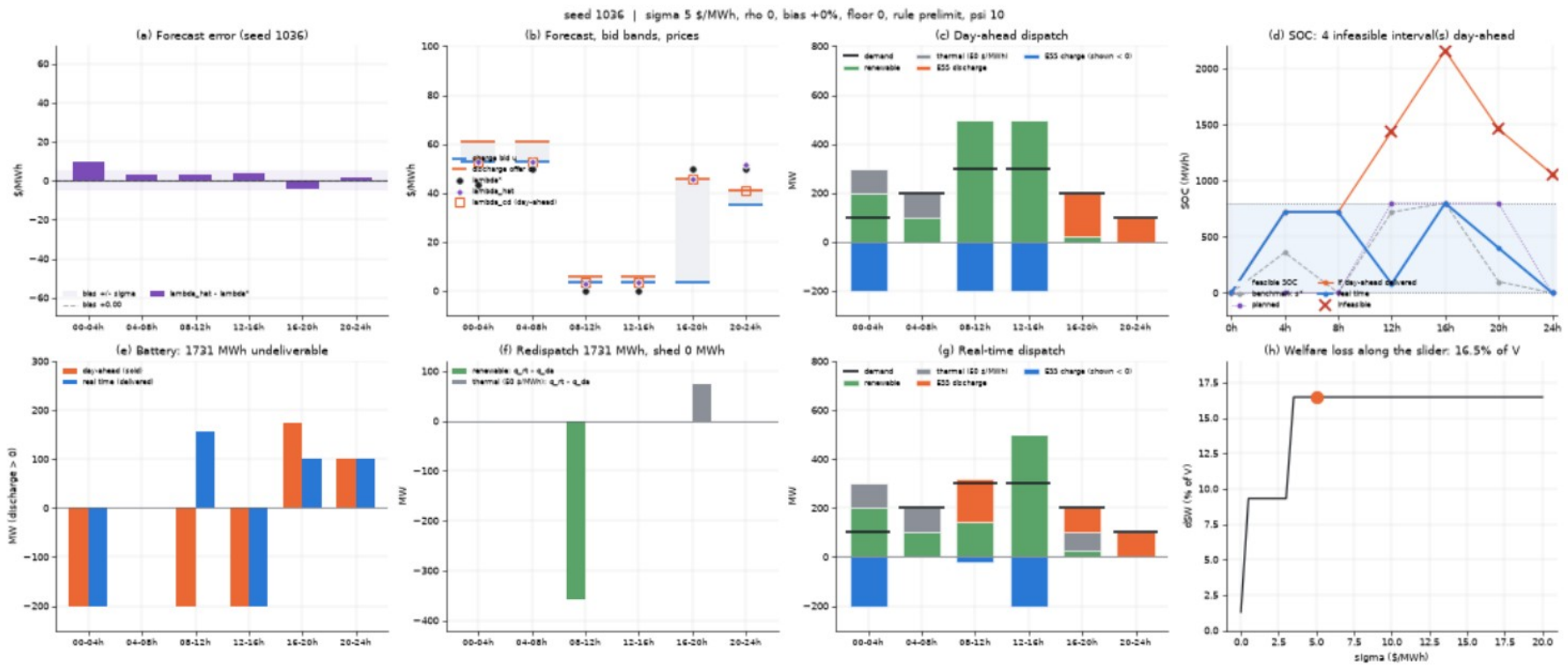


Reading: B★p − B★ = quantity plan · Bfcst − B★p = price forecast · Bfcst vs Bpriv = value of a forecast.

Same pipeline for all four: only the information behind the bid changes.

Final deliverable: every draw, every step, on one screen

5 · THE PIPELINE



One error shape (seed 1036) at σ = 5 $/MWh: the eight panels of the explorer. Panel (h) shows the loss along the σ slider.

Explorer: change one knob, all eight panels update

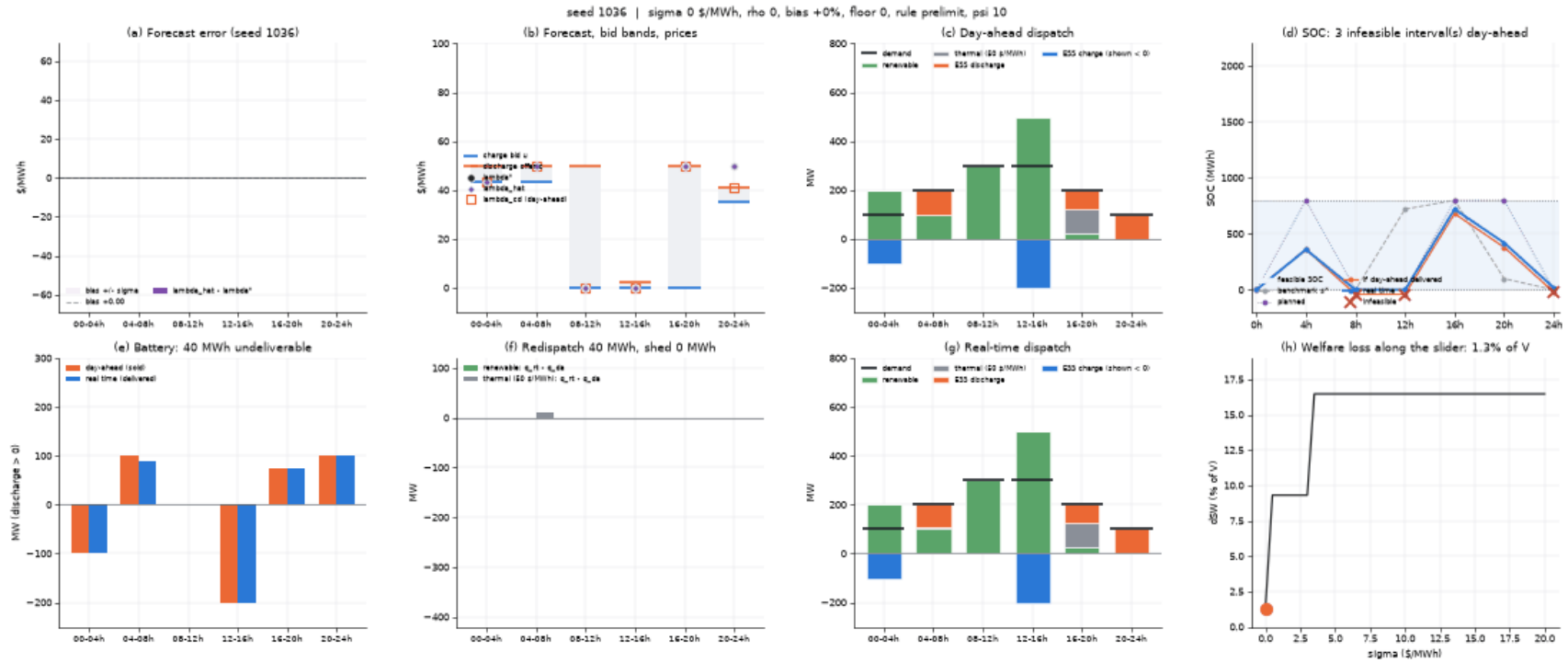
Albums: one slider per parameter, standalone HTML

The eight panels

- **a** forecast error
- **b** forecast, bid bands, prices
- **c** day-ahead dispatch
- **d** is the SOC feasible?
- **e** sold vs delivered
- **f** redispatch and shedding
- **g** real-time dispatch
- **h** where the loss comes from

Every number in the results can be traced back to a picture of the step that produced it.

Final deliverable: Sigma´s effect on the error model



One error shape (seed 1036) at σ = 5 $/MWh: the eight panels of the explorer. Panel (h) shows the loss along the σ slider.

Explorer: change one knob, all eight panels update

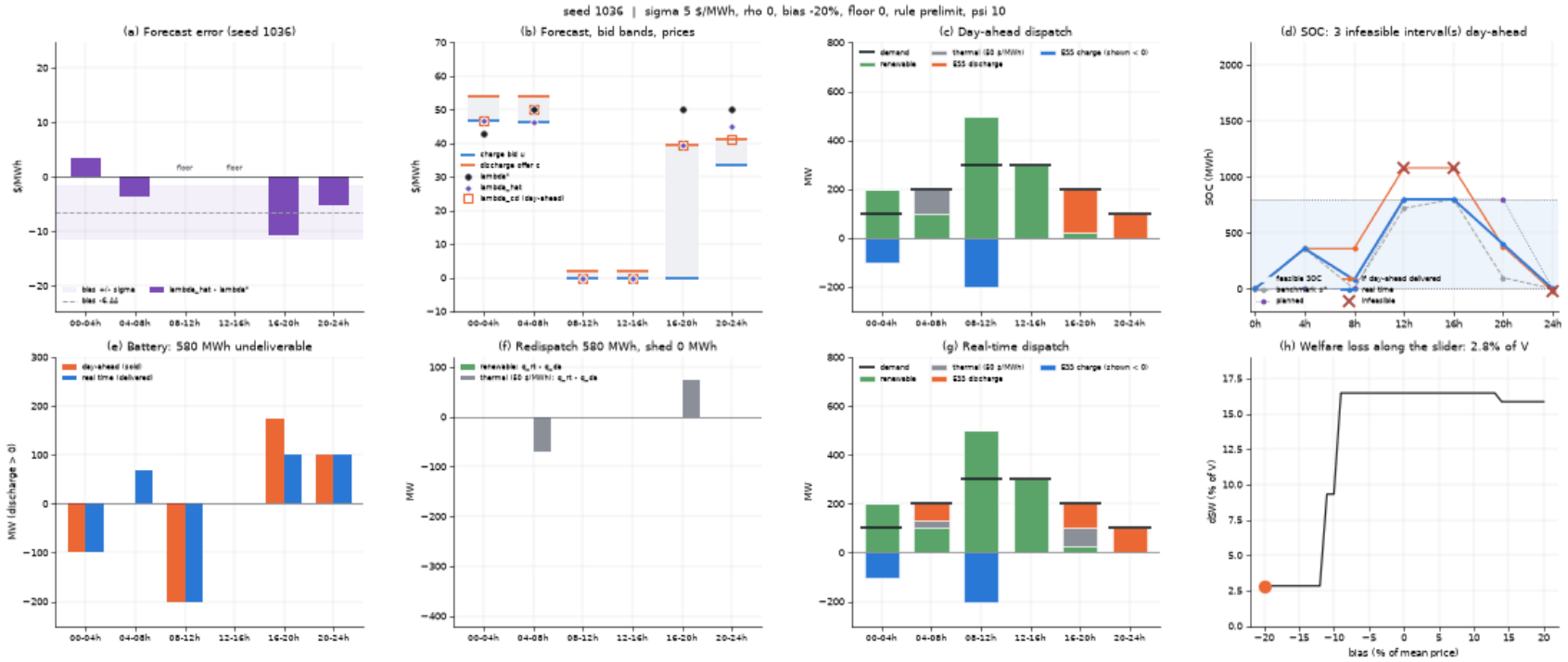
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Final deliverable: Bias´ effect on the error model



One error shape (seed 1036) at σ = 5 $/MWh: the eight panels of the explorer. Panel (h) shows the loss along the σ slider.

Explorer: change one knob, all eight panels update

Albums: one slider per parameter, standalone HTML

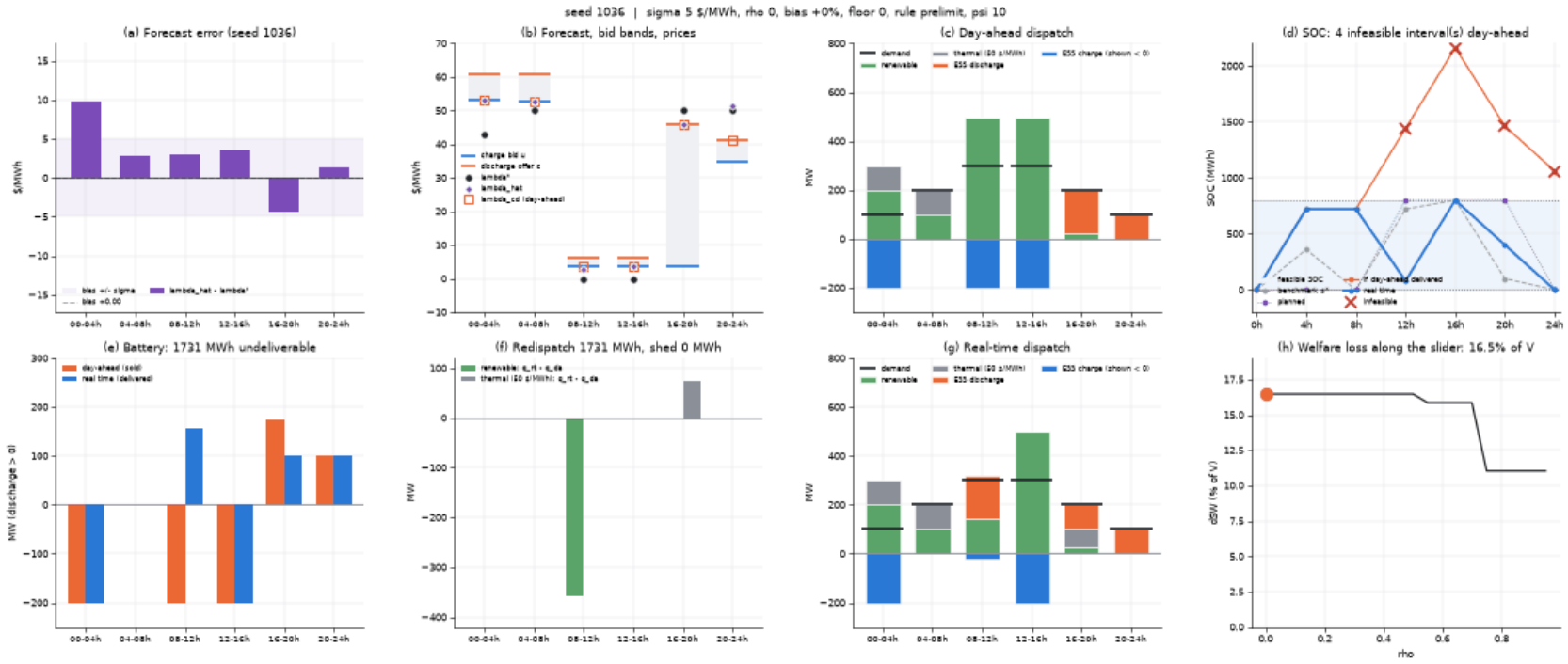
The eight panels

- **a** forecast error
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Every number in the results can be traced back to a picture of the step that produced it.

Final deliverable: Rho´s effect on the error model

5 · THE PIPELINE



One error shape (seed 1036) at σ = 5 $/MWh: the eight panels of the explorer. Panel (h) shows the loss along the σ slider.

Explorer: change one knob, all eight panels update

Albums: one slider per parameter, standalone HTML

The eight panels

- **a** forecast error
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DTU